SfM-FailSafe (synthetic validation)

Failure prediction and adaptive track weighting for sparse 3D reconstruction

Generated 2026-11-10 00:21 | Python 3.12.10 | COLMAP-based pipeline

WARNING: synthetic sphere scene and simulated mapper. Not COLMAP or ETH3D results.

Real COLMAP runs	0
Synthetic validation runs	16
Report artifacts	REPORT_SYNTHETIC.pdf

Synthetic pipeline validation

WARNING: generated sphere scene with a simulated mapper. Not COLMAP/ETH3D results; validation only.

run_name	cond	status	label_name	imgs	reg	points	mean_err
clean_s0	clean	success	success	12	12	2672	0.660
clean_s1	clean	success	success	12	12	2660	0.647
blur_mild_s0	blur_mild	success	success	12	12	2672	1.155
blur_mild_s1	blur_mild	success	success	12	12	2660	1.132
blur_severe_s0	blur_severe	success	failure	12	12	2672	2.309
blur_severe_s1	blur_severe	success	failure	12	12	2660	2.265
limited_views_s0	limited_views	success	success	6	6	2439	0.990
limited_views_s1	limited_views	success	success	6	6	2452	0.964
limited_views_severe_s0	limited_views_severe	success	failure	4	2	302	1.682
limited_views_severe_s1	limited_views_severe	success	success	4	4	1200	1.651
degraded_matches_mild_s0	degraded_matches_mild	success	success	12	12	2672	1.155
degraded_matches_mild_s1	degraded_matches_mild	success	success	12	12	2660	1.132
degraded_matches_severe_s0	degraded_matches_severe	success	failure	12	2	737	2.726
degraded_matches_severe_s1	degraded_matches_severe	success	failure	12	1	0	n/a
combined_severe_s0	combined_severe	success	failure	6	2	512	1.896
combined_severe_s1	combined_severe	success	failure	6	2	518	1.948

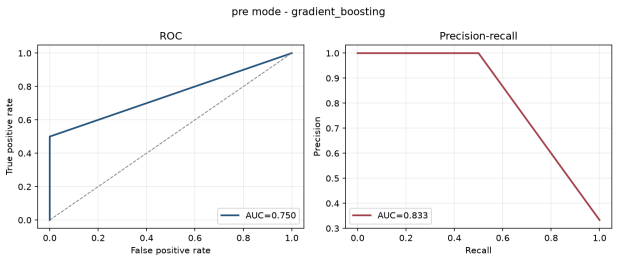
Synthetic failure model (pre)

Mode	pre
Leakage-free	True
Runs	16
Class counts	{'0': 9, '1': 7}
Features	38
Split	group_held_out
Best model	gradient_boosting
Threshold	0.5000
Preliminary	False

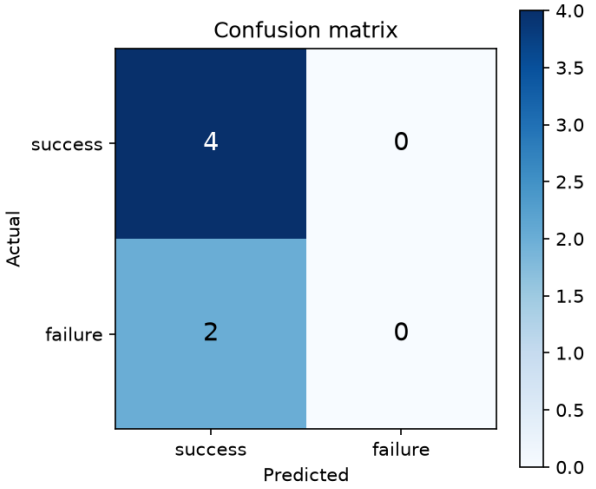
model	accuracy	precision	recall	f1	roc_auc	avg_prec	tp	tn	fp	fn
logistic_regression	0.667	0.000	0.000	0.000	0.625	0.500	0	4	0	2
random_forest	0.667	0.000	0.000	0.000	0.500	0.417	0	4	0	2
gradient_boosting	0.667	0.000	0.000	0.000	0.750	0.667	0	4	0	2

Held-out runs: blur_severe_s0, blur_severe_s1, clean_s0, clean_s1, limited_views_s0, limited_views_s1
Leave-one-condition-out (random forest):

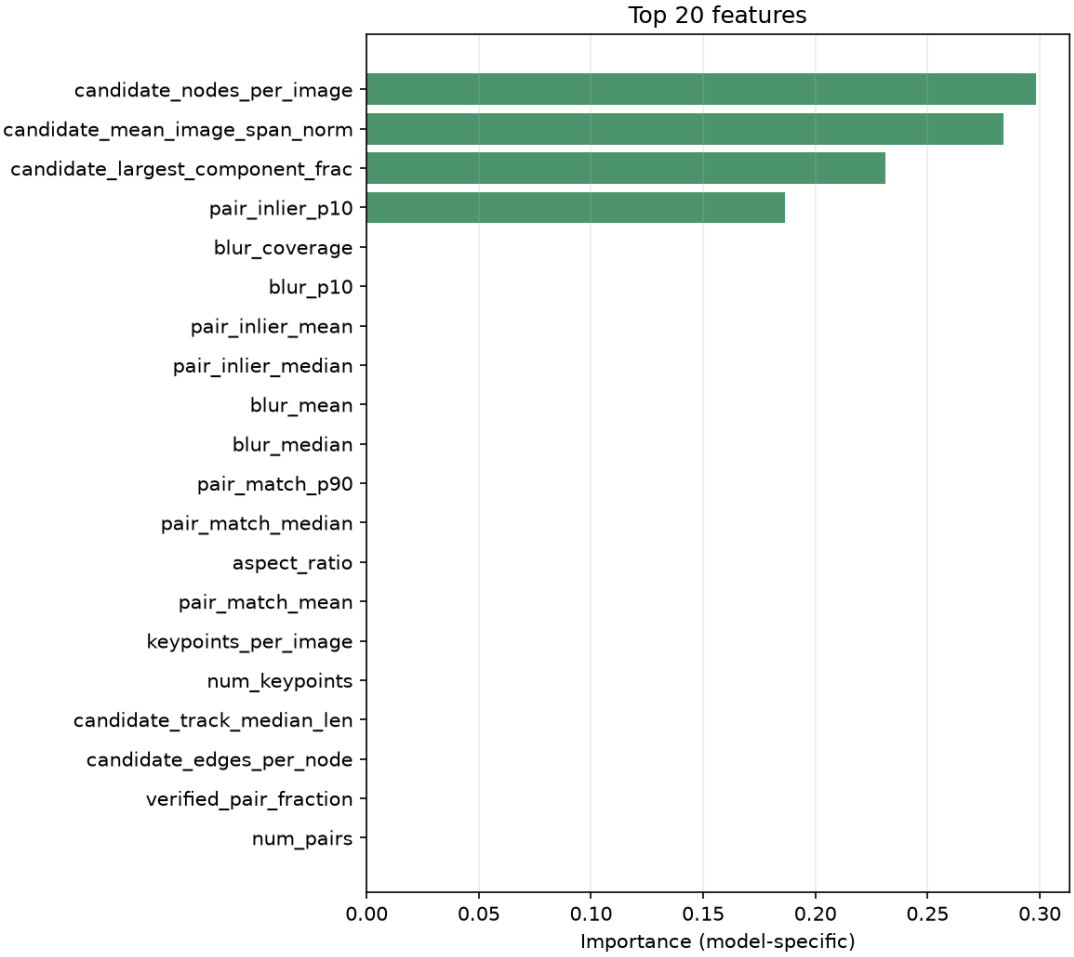
n	threshold	accuracy	precision	recall	f1	specificity	roc_auc	avg_prec	tp	tn	fp	fn	pred_pos	actual_pos
16.000	0.500	0.750	0.714	0.714	0.714	0.778	0.746	0.799	5.000	7.000	2.000	2.000	7.000	7.000



ROC / precision-recall



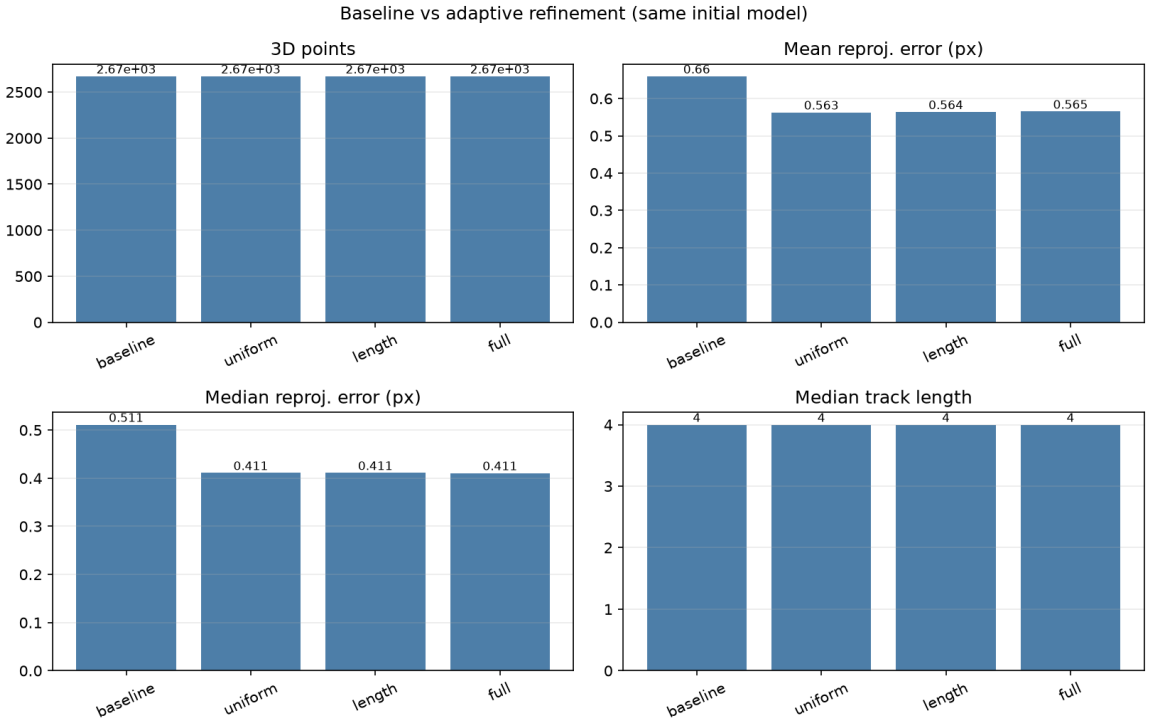
Confusion matrix



Feature importance

Synthetic adaptive comparison

variant	engine	points	mean_err	med_err	trk_med	success_label	sec
baseline	colmap_baseline	2672	0.660	0.511	4.000	0	2.147
uniform	scipy_least_squares	2672	0.563	0.411	4.000	0	9.900
length	scipy_least_squares	2672	0.564	0.411	4.000	0	8.533
full	scipy_least_squares	2672	0.565	0.411	4.000	0	8.596



Synthetic baseline vs refinement

Limitations

- Adaptive weighting is post-mapper refinement, not integrated into COLMAP's incremental mapper.
- scipy BA fixes intrinsics and gauge; it is not a Ceres/pyceres run.
- Failure labels and metrics are preliminary with few runs/conditions.
- Only ETH3D Courtyard and a synthetic scene are used; no cross-scene or ground-truth scan evaluation.
- pycolmap/pyceres were unavailable; COLMAP CLI conversion and scipy fallback were used.

Artifacts: outputs/runs, outputs/failure_model, outputs/experiments, outputs/REPORT.md, outputs/REPORT_SYNTHETIC.md